Introduction to Julia for High-Performance Computing

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Convenience

Performance

Language Barrier

Convenience

Performance

Gradual transition

	Tuesday	Wednesday	Thursday	Friday
	Foundations	**Core**	**Node**	**Cluster**
09:00 - 10:45	Getting Started	Type & Memory Optimizations	Multithreading	Distributed Computing
10:45 - 11:00	Break	Break	Break	Break
11:00 - 12:30	Fundamentals Exercises	Exercises	Exercises	Exercises
12:30 - 14:00	Lunch	Lunch	Lunch	Lunch
14:00 - 15:30	Compilation	SIMD / Profiling	GPU Computing	Q/A
15:30 - 15:45	Break	Break	Break	Outro
15:45 - 17:00	Exercises	Exercises	Exercises	

Quick Live Survey

Julia's Strengths

Julia is interactive
and convenient.

Julia knows maths and supports unicode.

```
σ(X, θ) = 1 ./ (1 .+ exp.(-X*θ))

predict(f, x) = f(x) > 0.5

function logistic_regression(X, y; T=1000, α=1e-4)
    X = mapreduce(x->[1;x]', vcat, X)
    θ = zeros(2)
    for iteration in 1:T
        θ = θ + α * X' * (y - σ(X,θ))
    end
    return x -> σ([1;x]', θ)
end
```

Julia has a great package manager

Laptop

```
→  ~/myproject  tree
.
├── Manifest.toml
├── Project.toml
└── code.jl

0 directories, 3 files

→  ~/myproject  cat Project.toml
[deps]
CUDA = "052768ef-5323-5732-b1bb-66c8b64840ba"
DifferentialEquations = "0c46a032-eb83-5123-abaf-570d42b7fbaa"
MKL = "33e6dc65-8f57-5167-99aa-e5a354878fb2"
MPI = "da04e1cc-30fd-572f-bb4f-1f8673147195"

→  ~/myproject  
```

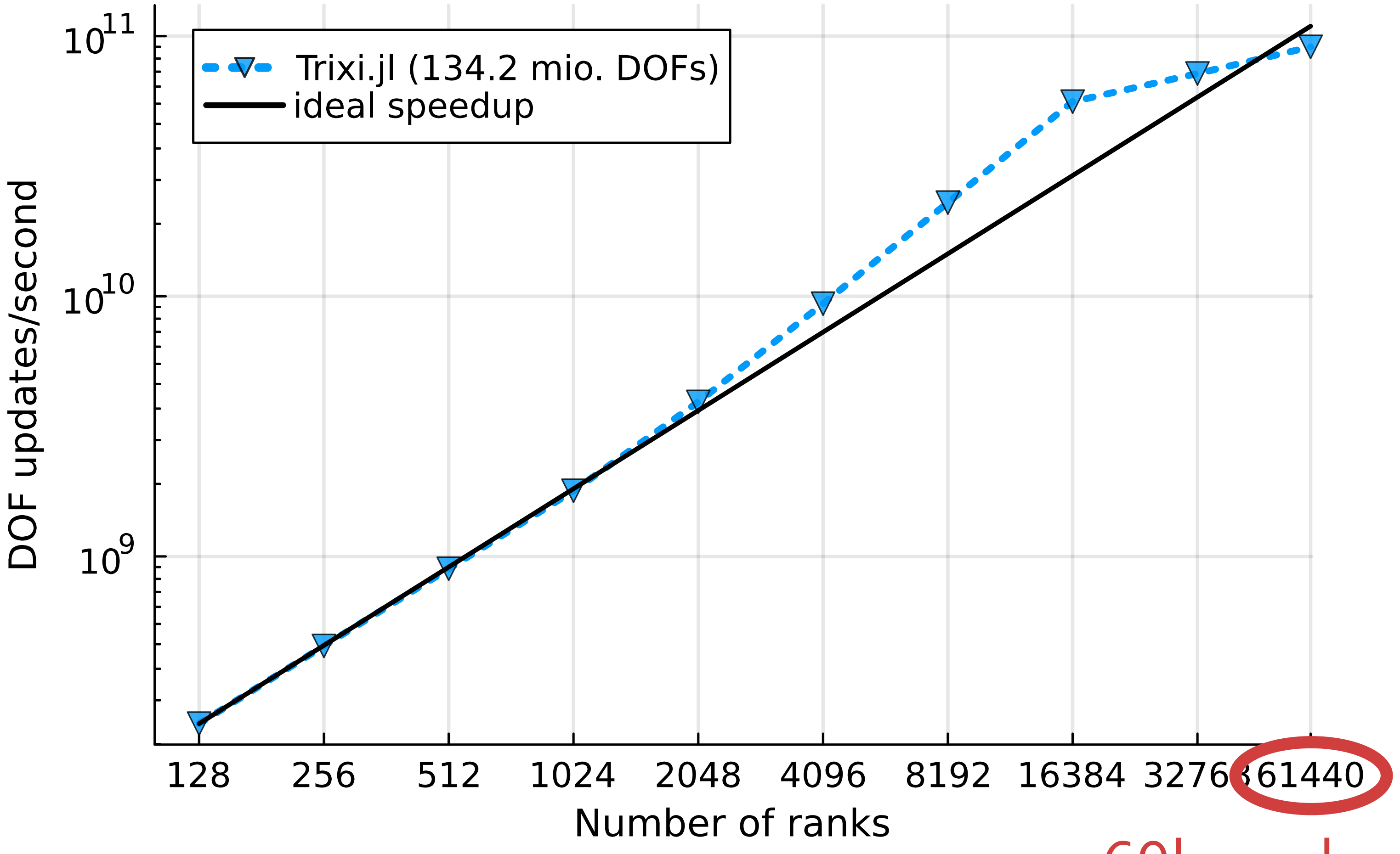
HPC Cluster

```
→  bauerc@n2login3 myproject  julia --project
               _
   _       _ _(_)_     |  Documentation: https://docs.julialang.org
  (_)     | (_) (_)    |
   _ _   _| |_  __ _   |  Type "?" for help, "]?" for Pkg help.
  | | | | | | |/ _` |  |
  | | |_| | | | (_| |  |  Version 1.7.2 (2022-02-06)
 _/ |\__'_|_|_|\__'_|  |  Official https://julialang.org/ release
|__/                   |

(myproject) pkg> st
      Status `~/myproject/Project.toml`
→ [052768ef] CUDA v3.11.0
→ [0c46a032] DifferentialEquations v7.1.0
→ [33e6dc65] MKL v0.5.0
→ [da04e1cc] MPI v0.19.2
        Info packages marked with → not downloaded, use `instantiat
e` to download

(myproject) pkg> instantiate
```

Julia code can be fast and scalable.

Example: Good scaling of PDE codes

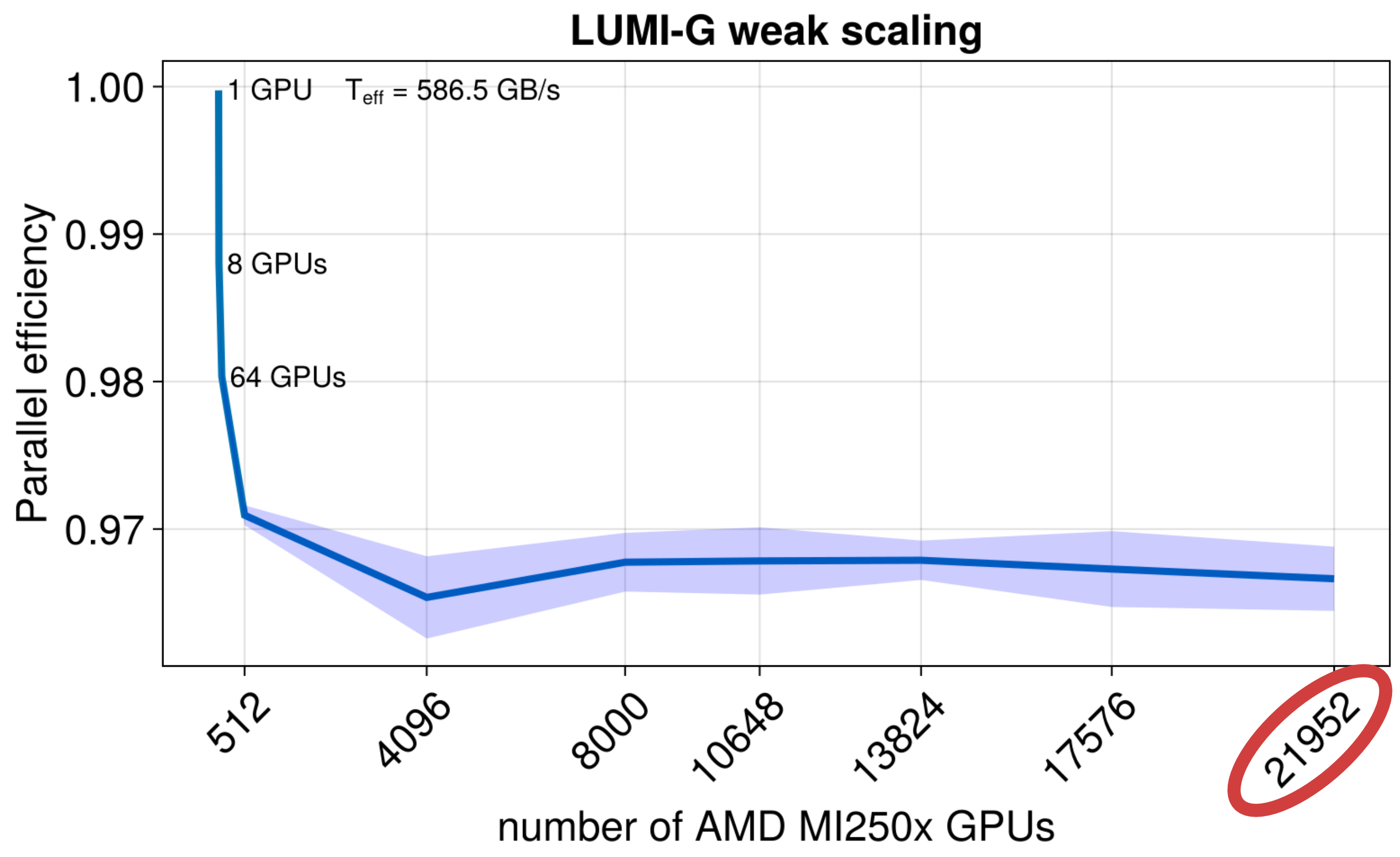
(higher is better)



≈ 60k ranks

Trixi.jl (Multi-CPU)

(higher is better)



≈ 22k GPUs

ParallelStencil.jl (Multi-GPU)

Julia is largely a **white box**.

Julia invites you to gradually dive deeper.

User

Developer

Gradual transition

Julia's Weaknesses

Julia can feel slow at times.

No great way to produce (small) binaries.

Julia is currently a **niche**.

Join us at JuliaCon ...

... or other conferences, like PASC, FOSDEM, or SC.

Achieving
high performance
can be tricky.

Let us delve deeper!